Integration: Definite Integrals and Areas

Questions 1–53

Question 1. Evaluate the following definite integrals:

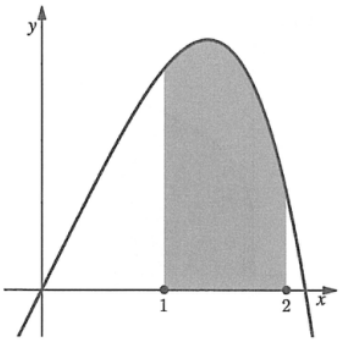
a) $\int_2^6 (8x+1)\,dx$

b) $\int_3^4 (6x^2-1)\,dx$

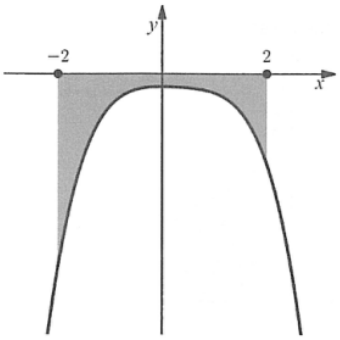
c) $\int_4^8 (500+4x-x^3)\,dx$

d) $\int_{-1}^1 (14x-9x^5)\,dx$

Question 2. The diagram below shows the curve with equation $y = 10x - x^4$. Find the exact value of the shaded area.

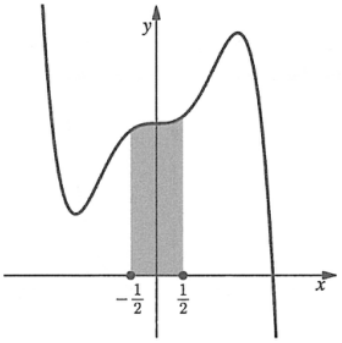


Question 3. The diagram below shows the curve with equation $y = -2 - x^2 + x^3 - x^4$. Find the exact value of the shaded area.



Question 4. It is given that $\int_1^3 (f(x) + 2)\,dx = 50$. Find the value of $\int_1^3 f(x)\,dx$.

Question 5. The diagram below shows the curve with equation $y = 10 - x^5 + 4x^3$. Find the exact value of the shaded area.

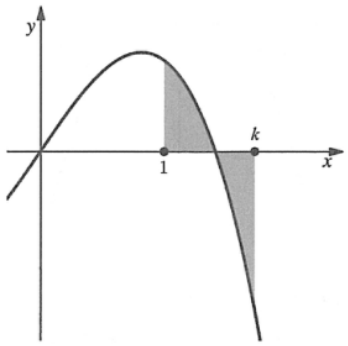


Question 6.

a) Sketch the graph of $y = (x - 1)(4 - x)$.

b) Explain why $\int_1^5 (x - 1)(4 - x)\,dx$ does not give the area between the curve, the x-axis and the lines $x = 1$ and $x = 5$.

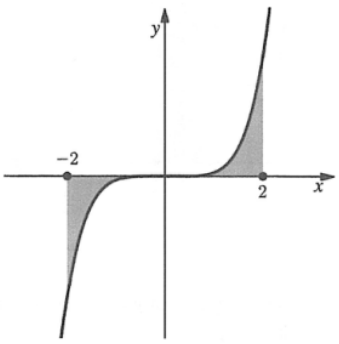
Question 7. The diagram below shows the curve with equation $y = 4x - 2x^3$. It is given that $\int_1^k (4x - 2x^3)\,dx = 0$ where k is a positive constant and $k > 1$. Find the value of k.



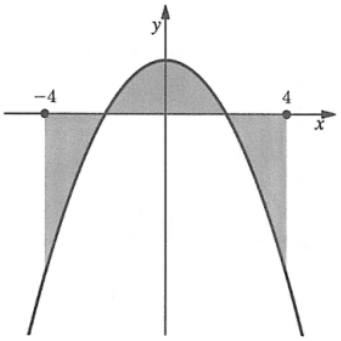
Question 8. A quadratic curve has the equation $y = ax^2 + bx + c$. It is given that the curve passes through $(-2, 0)$, $(6, 0)$ and $(0, 240)$. Find the exact finite area between the quadratic curve and the x-axis.

Question 9. Find the value of the constant k given that $\int_{-3}^{3} (kx^2 - 5x^4)\,dx = 18$.

Question 10. The diagram below shows the curve with equation $y = x^5$. Find the exact value of the shaded area.



Question 11. The diagram below shows the curve with equation $y = 4 - x^2$. Find the exact value of the shaded area.



Question 12. Show that $\int_{\sqrt{2}}^{\sqrt{3}} (3x^2 + 1)\, dx = 4\sqrt{3} - 3\sqrt{2}$.

Question 13. Show that $\int_1^{1+\sqrt{2}} (50 - 6x^2)\,dx = 40\sqrt{2} - 12$.

Question 14. Show that $\int_{1+\sqrt{3}}^{2+\sqrt{3}} 3x(x+2)\,dx = 25 + 15\sqrt{3}$.

Question 15. Find the exact value of the following definite integrals:

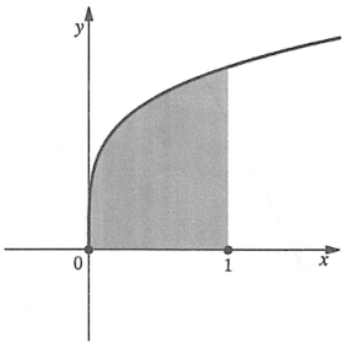
a) $\displaystyle\int_4^{16} \sqrt{x}\,dx$

b) $\displaystyle\int_{-2}^{-1} \frac{6}{x^2}\,dx$

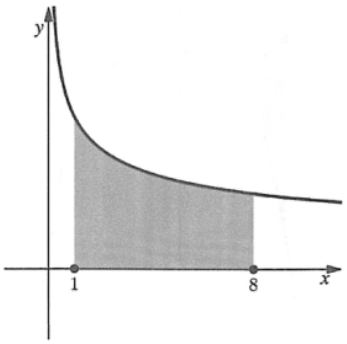
c) $\displaystyle\int_1^{9} \frac{4}{\sqrt{x}}\,dx$

d) $\displaystyle\int_1^{64} \left(\frac{1}{\sqrt[3]{x}} - \frac{1}{\sqrt[4]{x}}\right) dx$

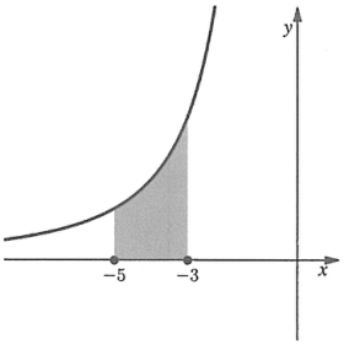
Question 16. The diagram below shows the curve with equation $y = \sqrt[4]{x}$. Find the exact value of the shaded area.



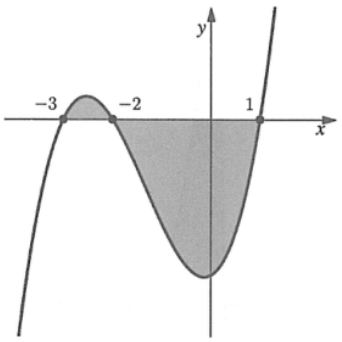
Question 17. The diagram below shows the curve with equation $y = \dfrac{12}{\sqrt[3]{x}}$. Find the exact value of the shaded area.



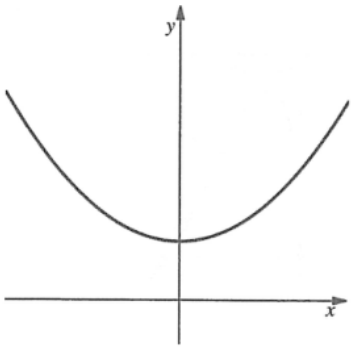
Question 18. The diagram below shows the curve with equation $y = \dfrac{9}{x^2}$. Find the exact value of the shaded area.



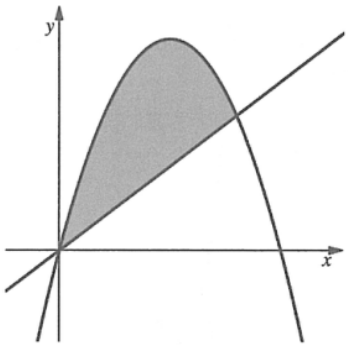
Question 19. The diagram below shows the curve with equation $y = x^3 + 4x^2 + x - 6$. Find the exact value of the shaded area.



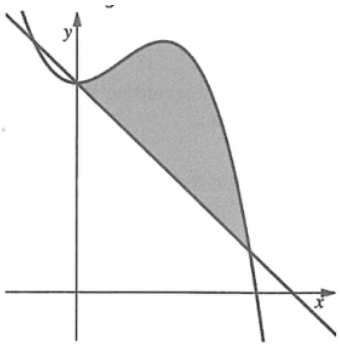
Question 20. The diagram below shows the curve with equation $y = \frac{1}{2}x^2 + 3$. Find the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 2k$, giving your answer in terms of k, where k is a non-zero constant.



Question 21. The diagram below shows the curve with equation $y = x(5 - x)$ and the line with equation $y = x$. Find the exact value of the shaded region.



Question 22. The diagram below shows the curve with equation $y = 20 - x^2(x - 3)$ and the line with equation $y = 20 - 4x$. Find the exact value of the shaded region.



Question 23. Find the value of the constant k given that $\int_1^2 \left(\frac{k}{x^3} + 2\right) dx = 5$.

Question 24. Find the value of the constant k given that $\int_1^9 (2kx - 3\sqrt{x})\, dx = 108$.

Question 25. Find the possible values of the constant k given that $\int_1^{16} (k\sqrt{x} + 2x + 2)\,dx = 3k^2$.

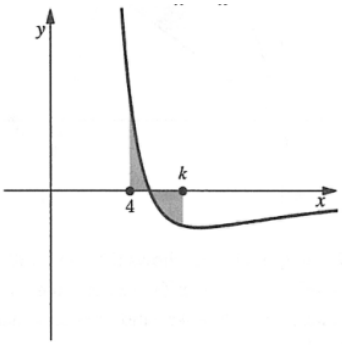
Question 26. Find the exact finite area enclosed between the curve $y = 6x^2 - 5x + 1$ and the x-axis.

Question 27. Find the maximum value of $\int_2^4 (3p - 2p^2x)\,dx$ where p is a positive constant.

Question 28. The diagram below shows the curve with equation $y = \frac{5}{x^3} - \frac{1}{x^2}$. It is given that

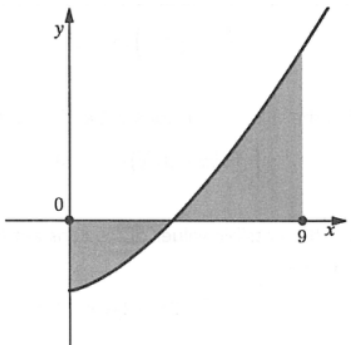
$$\int_4^k \left(\frac{5}{x^3} - \frac{1}{x^2}\right) dx = 0$$

where k is a positive constant with $k > 4$. Find the value of k.

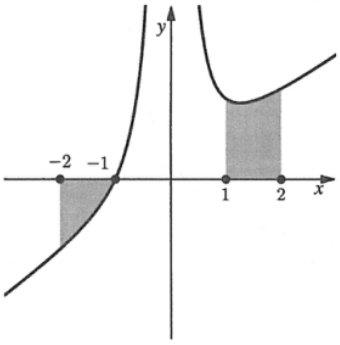


Question 29. Find the value of the constant k given that $\int_1^k \left(\frac{30}{\sqrt{x}} + 10\right) dx = 6090$.

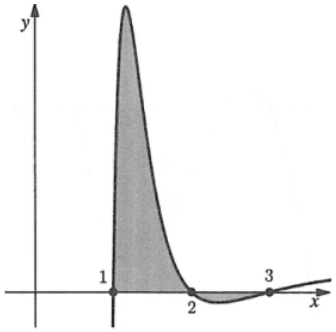
Question 30. The diagram below shows the curve with equation $y = x\sqrt{x} - 8$. Find the exact value of the shaded area.



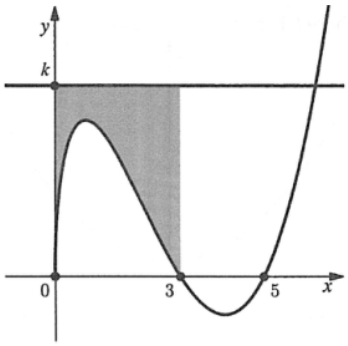
Question 31. The diagram below shows the curve with equation $y = x + \frac{1}{x^2}$. Find the exact value of the shaded area.



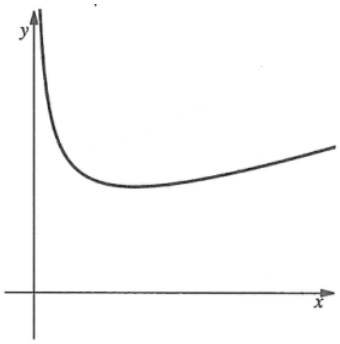
Question 32. The diagram below shows the curve with equation $y = \frac{1}{x^2} - \frac{6}{x^3} + \frac{11}{x^4} - \frac{6}{x^5}$. Find the exact value of the shaded area.



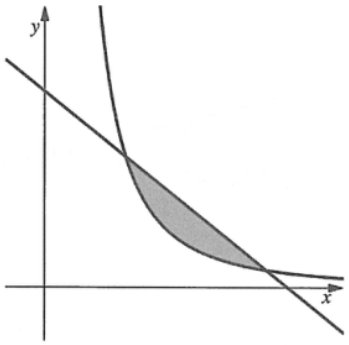
Question 33. The diagram below shows the curve with equation $y = 35(x-3)(x-5)\sqrt{x}$ and the line $y = k$. The shaded area is $300\sqrt{3}$ units^2. Find the value of k.



Question 34. The diagram below shows the curve with equation $y = \dfrac{2}{\sqrt{x}} + x$. Find the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 9k$, giving your answer in terms of k, where k is a positive constant.



Question 35. The diagram below shows the curve with equation $y = \dfrac{2}{x^2}$ and the line with equation $y = 3 - x$. Find the exact value of the shaded region.



Question 36. Find the exact value of the following definite integrals:

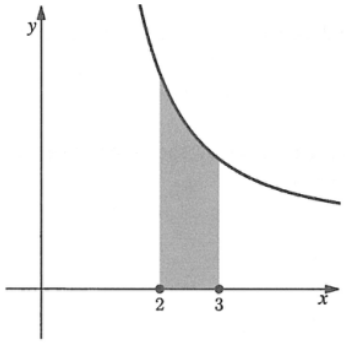
a) $\displaystyle\int_1^3 \frac{x+2}{x^3}\,dx$

b) $\displaystyle\int_2^4 \frac{3x+8}{x^4}\,dx$

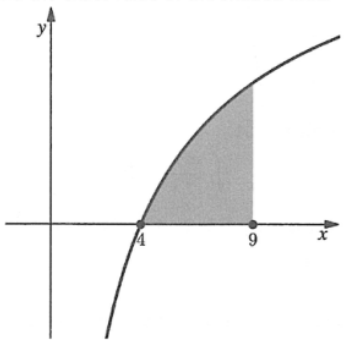
c) $\displaystyle\int_{-3}^{-1} \frac{4-5x}{2x^3}\,dx$

d) $\displaystyle\int_{-4}^{-2} \frac{3x^2-2}{6x^4}\,dx$

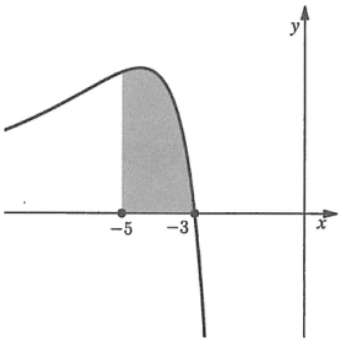
Question 37. The diagram below shows the curve with equation $y = \dfrac{x^2 + 10}{x^2}$. Find the exact value of the shaded area.



Question 38. The diagram below shows the curve with equation $y = \dfrac{\sqrt{x} - 2}{\sqrt{x}}$. Find the exact value of the shaded area.



Question 39. The diagram below shows the curve with equation $y = \dfrac{x+3}{x^3}$. Find the exact value of the shaded area.



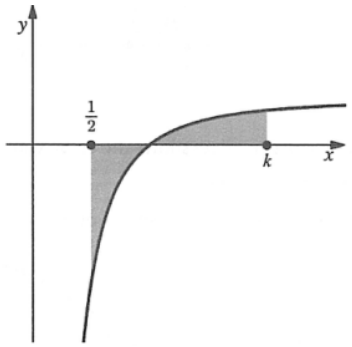
Question 40. Find the value of the constant k given that $\displaystyle\int_1^2 \frac{kx+1}{x^4}\,dx = \frac{11}{12}$.

Question 41. Find the value of the constant k given that $\displaystyle\int_4^9 \frac{4-k\sqrt{x}}{6\sqrt{x}}\,dx = 1$.

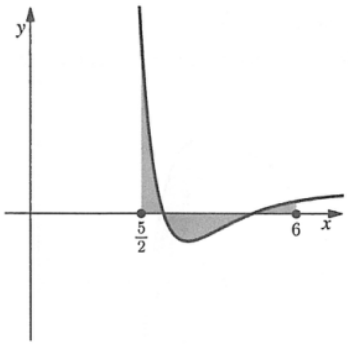
Question 42. The diagram below shows the curve with equation $y = \dfrac{x^2-1}{x^2}$. It is given that

$$\int_{1/2}^{k} \frac{x^2-1}{x^2}\,dx = 0$$

where k is a positive constant and $k > \dfrac{1}{2}$. Find the value of k.

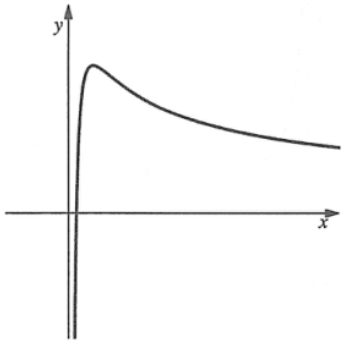


Question 43. The diagram below shows the curve with equation $y = \dfrac{(3-x)(5-x)}{x^4}$. Find the exact value of the shaded area.

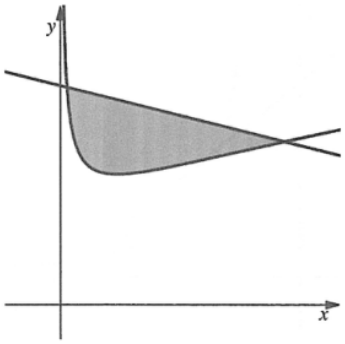


Question 44. Find the exact area enclosed between the curve $y = 84x^{5/2}$, the x-axis, and the line $x = 2$.

Question 45. The diagram below shows the curve with equation $y = \dfrac{3x-1}{x\sqrt{x}}$. Show that the area bounded by the curve, the x-axis, and the lines $x = k$ and $x = 4k$, where k is a positive constant, is $\dfrac{6k-1}{\sqrt{k}}$.



Question 46. The diagram below shows the curve with equation $y = \dfrac{x+1}{\sqrt{x}}$ and the line with equation $3x+14y = 47$. Find the exact value of the shaded region.

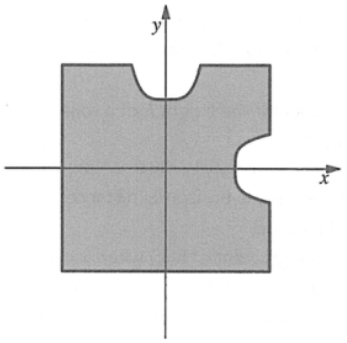


Question 47. Prerequisite knowledge for Q47: Graph Sketching: Radical Functions
Find the exact finite area enclosed between the curve $y = \sqrt{x} - 16$, the x-axis and the y-axis.

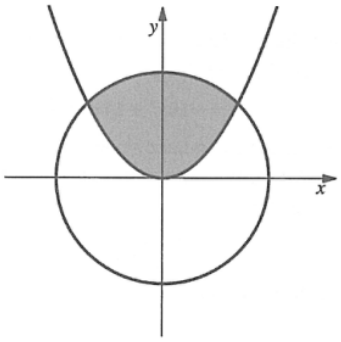
Question 48. Prerequisite knowledge for Q48: Graph Sketching: Cubic Graphs & Quartic Graphs
Find the exact finite area enclosed between the following curves and the x-axis:

a) $y = (x-1)(x-2)(x-3)$
b) $y = (x-1)^2(x-2)$
c) $y = (x-1)(x-2)^2$
d) $y = (x-1)^2(x-2)^2$

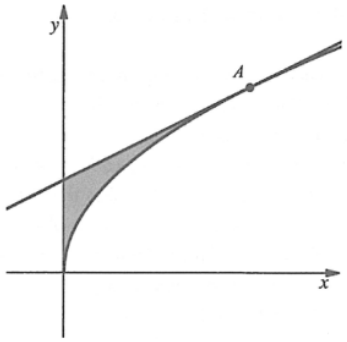
Question 49. A puzzle piece is designed using a 6 cm by 6 cm square with the origin as its centre, as shown in the diagram below. Two identical curved shapes are cut out of the square to allow other pieces to be joined on the sides. The designer uses the equation $y = x^4 + 2$ to create one of the curved sections. Find the exact area of the puzzle piece.



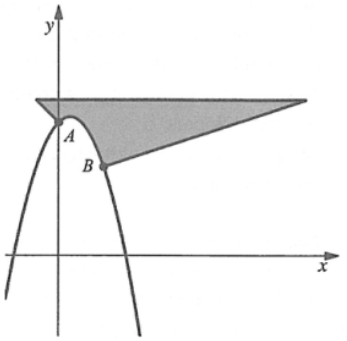
Question 50. The diagram below shows the circle with equation $x^2 + y^2 = 2$ and the curve with equation $y = x^2$. Find the exact value of the shaded area.



Question 51. Prerequisite knowledge for Q51–Q53: Applications of Differentiation
The diagram below shows the curve with equation $y = 2\sqrt{x}$. The tangent to the curve at $x = 4$ is also shown in the diagram. The point A is where the tangent meets the curve. The shaded area is the region bounded by the curve, the tangent and the y-axis. Find the exact value of the shaded area.



Question 52. The diagram below shows the curve with equation $y = (3 - x)(x + 2)$. The points A and B lie on the curve and have x-coordinates 0 and 2 respectively. The shaded region is formed by the curve, the line $y = 7$, the normal to the curve at A and the normal to the curve at B. Find the exact shaded area.



Question 53. The diagram below shows the curve with equation $y = 2x^3 + 15x^2 - 144x + 500$. The points A and B lie on the turning points of the curve. Find the exact value of the shaded area.

